

SBP ATCT ROOF REPLACEMENT



**WESTERN SERVICE AREA
ENGINEERING SERVICES
AWP**

**STATEMENT OF WORK
ROOF REPLACEMENT
SAN LUIS OBISPO, CALIFORNIA**

SBP ATCT ROOF REPLACEMENT

SOW & SPECIFICATIONS FOR REPLACING ROOFING AT ATCT San Luis Obispo, California

PROJECT LOCATION:

San Luis Obispo Municipal Airport

925 Airport Drive

San Luis Obispo, California 93401

1- BASE SCOPE:

The Subcontractor shall furnish all labor, materials, equipment, supervision and all other services required to replace the roof at ATCT at SBP. Attached drawings are for reference only, site visit encouraged. Approximate quantities ATCT Cab roof 361 sq. ft.

The scope of work includes but not limited to:

- Remove and dispose of existing roofing system from the SBP ATCT cab roof.
- Provide and install roof hatch guardrail kit with self closing gate, galvanized finish. to meet code compliance.
- Remove and replace all roof drains.
- Remove and replace roof hatch
- Remove and relocate grounding counterpoise from roof to parapet wall.
- Remove debris and trash generated from work and dispose of the debris per local and state requirements, daily.

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All safety, security, and protections rules and regulation required by the federal, state and local entities shall be followed and adhered to. Any deviation from these rules may result in work stoppage by the FAA COR.

MATERIAL OPTIONS:

The material option shall be for entire roof areas to meet the specifications listed at the end of this document in Section 07 54 23 – Thermoplastic Polyolefin (TPO) roofing:

2- INSTALLATION DIRECTIONS:

The roofing material used for the Air Traffic Control Tower shall follow the recommended manufacturers' recommendations and Section 07 54 23.

3- SITE RESTRICTION:

Work hours will be restricted from 0700 – 1600 and require coordination with local air traffic to have the least impact on the NAS. All contractors and subcontractors shall be accompanied by the onsite COR.

4- SUBMITTALS:

The subcontractor shall provide to Parsons one week after the award of contract the following documents.

1. Project Health & Safety Plan.
2. Product Safety Data Sheets.
3. List of all employees on site.

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5- SAFETY REQUIREMENTS:

The contractor is solely responsible for the safety and welfare of workers on the project and the public around the construction site. Contractor shall take precautions to adequately safeguard the conditions of all personal on or near the site. Comply with the regulations of the Occupational Safety and Health Standards (OSHA), 29 CFR Part 1926. If COR deems that the contractor is not safe, he/she has the right to stop all work until the contractor corrects the safety violation. contractor shall submit a Health and Safety Plan for this project. The Health and Safety Plan shall identify the competent safety individual for the project to monitor safety and identify safety hazards.

All existing equipment and materials removed from the project area shall become property of the contractor. contractor shall adhere to local government disposal regulations.

Contractors shall maintain the job site in a neat and orderly condition; this includes the daily removal of rubbish, waste and tools, equipment, and materials not required for the work in progress.

6- SCHEDULING:

The contractor shall submit a project schedule after award.

7- PERFORMANCE TIME:

Work shall start after notice to proceed (NTP) has been issued, and be done diligently and completed in no later than 30 calendar days. The working hours should be M-F from 7:00 AM to 4:00 PM, in coordination with air traffic. No work shall be scheduled for the weekends and official holidays unless approved by Contracting Officer in advance, with no additional cost to government.

The subcontractor shall confine operations, activities, material storage, and employee parking within the areas shown him COR.

8- GOVERNMENT FURNISHED MATERIAL:

NONE

9- EXTRA MATERIALS:

NONE.

10- EMERGENCY CONTACT LIST:

The contractor shall provide an emergency contact list of names and telephone numbers for at least the foreman and a backup person so that contact can be made during non-working hours in case there is an emergency situation.

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11- PROJECT SUPERVISION:

The contractor shall have a designated project superintendent at the project site at all times during the course of work. A FAA RE will be present for supervision as well.

12- AVAILABILITY OF UTILITIES:

Contractor to provide all utilities to complete the job.

13- PRE CONSTRUCTION MEETING:

Pre-construction meeting will be scheduled prior to start of the work to discuss and respond to any questions the subcontractor may have – and assure the contractor adheres to all technical security and safety requirements. Use of the facility restrooms, water, and power will also be discussed. Compliance with these procedures while on site is mandatory. The date and time for the pre-conference meeting will be determined by the CO at later time.

14- INSPECTIONS:

COR will inspect all work in progress up to completion and final Contractor's Acceptance Inspection (CAI) including all materials, tools, and equipment. Such inspection may extend to all or a part of the work for the preparation, and materials to be used. When contractors are notified by the COR of any non-compliance with the provisions of the contract and the action(s) to be taken, the contractor shall correct the unsafe or unhealthy condition. The CO may stop the work with a "stop work order" if the contractor fails to comply promptly with all or any part of the work being performed. When satisfactory corrective action has been taken to correct the unsafe and unhealthy condition, the contractor may resume work when approved by the CO. The Contractor shall not be allowed any extension of time or compensation for damages by reason of or in connection with such work stoppage.

15- PROTECTION OF EXISTING FACILITY:

The contractor shall take all precautions necessary to protect the existing facilities, equipment, buildings, vegetation, etc., during construction. Any areas damaged by the contractor or any tier subcontractors, shall be repaired or replaced to their original conditions by the contractor at no additional cost to the government. Repairs shall be approved by the RE and shall match to original finish. The contractor shall provide all temporary covers, enclosures, barricades, etc., required to protect the existing facilities.

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16- CLEAN UP

Proper clean up and removal of the debris from the job site, including any hazardous waste produced by any paint or construction materials used. The work site shall be kept clean and orderly during the progress of work. After final inspection, but before final acceptance of the work, all exterior surfaces of the equipment shall be cleaned of dirt, mud, or oil accumulations. All debris shall be removed from the general site area.

17- WARRANTIES:

Documentation for the warranties shall be provided to FAA upon completion of the project. Warranty period shall be a minimum of 5 years from the date of the Final Acceptance by the Government. Warranty shall include repair or replacement of carpet that does not meet the requirements in this scope of work or fails in materials or workmanship. Failures include, but are not limited to, more than 10% loss of face fiber, edge raveling, snags, runs, and delamination.

18- APPENDIX:

04 54 23 TPO Roofing

07 72 00 accessories

END OF DOCUMENT

SECTION 07 54 23 - THERMOPLASTIC POLYOLEFIN (TPO) ROOFING

PART 1 - GENERAL

SUMMARY

A. Section Includes:

1. Mechanically fastened TPO membrane roofing system.
2. Vapor retarder.
3. Roof insulation.

1.2 DEFINITIONS

- A. TPO: Thermoplastic polyolefin.
- B. Roofing Terminology: See ASTM D 1079 and glossary in NRCA's "The NRCA Roofing and Waterproofing Manual" for definitions of terms related to roofing work in this Section.

1.3 PERFORMANCE REQUIREMENTS

- A. General Performance: Installed membrane roofing and base flashings shall withstand specified uplift pressures, thermally induced movement, and exposure to weather without failure due to defective manufacture, fabrication, installation, or other defects in construction. Membrane roofing and base flashings shall remain watertight.
- B. Material Compatibility: Provide roofing materials that are compatible with one another under conditions of service and application required, as demonstrated by membrane roofing manufacturer based on testing and field experience.
- C. FM Approvals Listing: Provide membrane roofing, base flashings, and component materials that comply with requirements in FM Approvals 4450 and FM Approvals 4470 as part of a membrane roofing system, and that are listed in FM Approvals' "RoofNav" for Class 1 or noncombustible construction, as applicable. Identify materials with FM Approvals markings.
1. Fire/Windstorm Classification: Class 1A-135.
 2. Hail Resistance: SH.
- D. Energy Performance: Provide roofing system that is listed on the DOE's ENERGY STAR "Roof Products Qualified Product List" for low-slope roof products. Roofing system shall meet R-30 insulation factor.

1.4 SUBMITTALS

- A. Product Data: For each type of product indicated.

1. Adhesives: Provide VOC documentation.
- B. Shop Drawings: For roofing system. Include plans, elevations, sections, details, and attachments to other work.
 1. Base flashings and membrane terminations.
 2. Tapered insulation, including slopes.
 3. Roof plan showing orientation of steel roof deck and orientation of membrane roofing and fastening spacings and patterns for mechanically fastened membrane roofing.
 4. Insulation fastening patterns for corner, perimeter, and field-of-roof locations.
- C. Qualification Data: For qualified Installer, names of architects and owners, manufacturer and inspector.
- D. Manufacturer Certificates: Signed by roofing manufacturer certifying that roofing system complies with requirements specified in "Performance Requirements" Article.
 1. Submit evidence of compliance with performance requirements.
- E. Product Test Reports: Based on evaluation of comprehensive tests performed by manufacturer and witnessed by a qualified testing agency, for components of membrane roofing system.
- F. Research/Evaluation Reports: For components of membrane roofing system, from the ICC-ES.
- G. Field quality-control reports.
- H. Maintenance Data: For roofing system to include in maintenance manuals.
- I. Warranties: Sample of special warranties.

1.5 QUALITY ASSURANCE

- A. Manufacturer Qualifications: A qualified manufacturer that is UL listed and FM Approvals approved for membrane roofing system identical to that used for this Project.
- B. Installer Qualifications: A qualified firm that is approved, authorized, or licensed by membrane roofing system manufacturer to install manufacturer's product and that is eligible to receive manufacturer's special warranty.
- C. Inspector Qualifications: Provide qualifications of a roofing consultant who shall be responsible for field quality control. Inspector shall be an independent testing and field quality control professional with a minimum 5 years experience in similar project size, complexity and system to conduct testing indicated.
- D. Source Limitations: Obtain components including roof insulation and fasteners for membrane roofing system from same manufacturer as membrane roofing or approved by membrane roofing manufacturer.

- E. Exterior Fire-Test Exposure: ASTM E 108, Class A; for application and roof slopes indicated, as determined by testing identical membrane roofing materials by a qualified testing agency. Materials shall be identified with appropriate markings of applicable testing agency.
- F. Fire-Resistance Ratings: Where indicated, provide fire-resistance-rated roof assemblies identical to those of assemblies tested for fire resistance per ASTM E 119 by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.
- G. Preliminary Roofing Conference: Before starting roof deck construction, conduct conference at Project site.
 - 1. Meet with FAA, COTR, Government's insurer if applicable, testing and inspecting agency representative, roofing installer, roofing system manufacturer's representative, deck Installer, and installers whose work interfaces with or affects roofing, including installers of roof accessories and roof-mounted equipment.
 - 2. Review methods and procedures related to roofing installation, including manufacturer's written instructions.
 - 3. Review and finalize construction schedule and verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.
 - 4. Review deck substrate requirements for conditions and finishes, including flatness and fastening.
 - 5. Review structural loading limitations of roof deck during and after roofing.
 - 6. Review base flashings, special roofing details, roof drainage, roof penetrations, equipment curbs, and condition of other construction that will affect roofing system.
 - 7. Review governing regulations and requirements for insurance and certificates if applicable.
 - 8. Review temporary protection requirements for roofing system during and after installation.
 - 9. Review roof observation and repair procedures after roofing installation.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Deliver roofing materials to Project site in original containers with seals unbroken and labeled with manufacturer's name, product brand name and type, date of manufacture, approval or listing agency markings, and directions for storing and mixing with other components.
- B. Store liquid materials in their original undamaged containers in a clean, dry, protected location and within the temperature range required by roofing system manufacturer. Protect stored liquid material from direct sunlight.
 - 1. Discard and legally dispose of liquid material that cannot be applied within its stated shelf life.
- C. Protect roof insulation materials from physical damage and from deterioration by sunlight, moisture, soiling, and other sources. Store in a dry location. Comply with insulation manufacturer's written instructions for handling, storing, and protecting during installation.
- D. Handle and store roofing materials and place equipment in a manner to avoid permanent deflection of deck.

1.7 PROJECT CONDITIONS

- A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit roofing system to be installed according to manufacturer's written instructions and warranty requirements.

1.8 WARRANTY

- A. Special Warranty: Manufacturer's standard or customized form, without monetary limitation, in which manufacturer agrees to repair or replace components of membrane roofing system that fail in materials or workmanship within specified warranty period.
 - 1. Special warranty includes membrane roofing, base flashings, roof insulation, fasteners, cover boards, roofing accessories, and other components of membrane roofing system.
 - 2. Warranty for roofing system shall be a no dollar limit for labor and materials.
 - 3. Warranty Period: 20 years from date of Substantial Completion.
- B. Puncture Resistance Warranty: In addition to Special Warranty, Contractor agrees to repair and replace all or part of components of membrane roofing system caused by accidental punctures. Provide a no dollar limit, 20-year warranty.
- C. One-Year Warranty Inspection: As part of the one-year warranty inspection, the COTR will conduct an infrared roof survey on any project involving a membrane roofing system. This survey will be conducted in accordance with ASTM C 1153. The Contractor shall be required to replace all damaged materials and to locate and repair sources of moisture penetration, at no additional cost to the FAA.

PART 2 - PRODUCTS

2.1 TPO MEMBRANE ROOFING

- A. Fabric-Reinforced Thermoplastic Polyolefin Sheet: ASTM D 6878, internally fabric or scrim reinforced, uniform, flexible TPO sheet.
 - 1. Manufacturers: Subject to compliance with requirements, provide products by Stevens Roofing Systems; Division of JPS Elastomerics. Other available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Carlisle SynTec Incorporated.
 - 2. Thickness: 80 mils, nominal with a minimum of 34 mils above scrim.
 - 3. Exposed Face Color: White.

2.2 AUXILIARY MEMBRANE ROOFING MATERIALS

- A. General: Auxiliary membrane roofing materials recommended by roofing system manufacturer for intended use, and compatible with membrane roofing.
- B. Sheet Flashing: Manufacturer's standard unreinforced thermoplastic polyolefin sheet flashing, 45 mils thick, minimum, of same color as sheet membrane. Use 60 mil thick unreinforced thermoplastic polyolefin sheet flashing at parapet, same color as sheet membrane.
- C. Bonding Adhesive: Manufacturer's standard, water based. Adhesive use will be confirmed by COTR and shall have low odor.
- D. Slip Sheet: Manufacturer's standard, of thickness required for application.
- E. Metal Termination Bars: Manufacturer's standard, predrilled stainless-steel or aluminum bars, approximately 1 by 1/8 inch thick; with anchors.
- F. Metal Battens: Manufacturer's standard, aluminum-zinc-alloy-coated or zinc-coated steel sheet, approximately 1 inch wide by 0.05 inch thick, prepunched.
- G. Fasteners: Factory-coated steel fasteners and metal or plastic plates complying with corrosion-resistance provisions in FM Approvals 4470, designed for fastening membrane to substrate, and acceptable to membrane roofing system manufacturer.
- H. Miscellaneous Accessories: Provide pourable sealers, preformed cone and vent sheet flashings, preformed inside and outside corner sheet flashings, T-joint covers, lap sealants, termination reglets, and other accessories.

2.3 SUBSTRATE BOARDS

- A. Substrate Board: ASTM C 1177/C 1177M, glass-mat, water-resistant gypsum substrate, Type X, 5/8 inch thick.
 - 1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Georgia-Pacific Corporation; Dens Deck.
 - 2. Substrate boards shall be placed directly on metal deck with fasteners.
- B. Fasteners: Factory-coated steel fasteners and metal or plastic plates complying with corrosion-resistance provisions in FM Approvals 4470, designed for fastening substrate board to roof deck.

2.4 PROTECTION BOARDS

- A. Protection Board: ASTM C 1177/C 1177M, glass-mat, water-resistant gypsum substrate, Type X, 1/2 inch thick.

1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
 - a. Georgia-Pacific Corporation; Dens Deck.
2. Protection board shall be placed on top of rigid insulation; roof membrane shall be placed directly on protection board forming roof assembly.

2.5 VAPOR RETARDER

- A. Polyethylene Film: ASTM D 4397, 6 mils thick, minimum, with maximum permeance rating of 0.13 perm.
 1. Tape: Pressure-sensitive tape of type recommended by vapor-retarder manufacturer for sealing joints and penetrations in vapor retarder.

2.6 ROOF INSULATION

- A. General: Preformed roof insulation boards manufactured or approved by TPO membrane roofing manufacturer, selected from manufacturer's standard sizes suitable for application, of thicknesses indicated and that produce FM Approvals-approved roof insulation.
- B. Polyisocyanurate Board Insulation: ASTM C 1289, Type II, Class 1, Grade 2, felt or glass-fiber mat facer on both major surfaces.
- C. Provide preformed saddles, crickets, tapered edge strips, and other insulation shapes where indicated for sloping to drain. Fabricate to slopes indicated.

2.7 INSULATION ACCESSORIES

- A. General: Furnish roof insulation accessories recommended by insulation manufacturer for intended use and compatibility with membrane roofing.
- B. Fasteners: Factory-coated steel fasteners and metal or plastic plates complying with corrosion-resistance provisions in FM Approvals 4470, designed for fastening roof insulation to substrate, and acceptable to roofing system manufacturer.

2.8 WALKWAYS

- A. Flexible Walkways: Factory-formed, nonporous, heavy-duty, slip-resisting, surface-textured walkway pads or rolls, approximately 3/16 inch thick, and acceptable to membrane roofing system manufacturer.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, for compliance with the following requirements and other conditions affecting performance of roofing system:
 - 1. Verify that roof openings and penetrations are in place and curbs are set and braced and that roof drain bodies are securely clamped in place.
 - 2. Verify that wood blocking, curbs, and nailers are securely anchored to roof deck at penetrations and terminations and that nailers match thicknesses of insulation.
 - 3. Verify that metal deck is acceptable for installation of roofing system.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

- A. Clean substrate of dust, debris, moisture, and other substances detrimental to roofing installation according to roofing system manufacturer's written instructions. Remove sharp projections.
- B. Prevent materials from entering and clogging roof drains and conductors and from spilling or migrating onto surfaces of other construction. Remove roof-drain plugs when no work is taking place or when rain is forecast.
- C. Complete terminations and base flashings and provide temporary seals to prevent water from entering completed sections of roofing system at the end of the workday or when rain is forecast. Remove and discard temporary seals before beginning work on adjoining roofing.
- D. Close all fresh air intakes/louvers where dust and odors from the roofing activities could enter the building.
- E. Coordinate with the COTR the timing of the roofing removals and installation activities with the least potential for disruption of building operations.

3.3 SUBSTRATE BOARD

- A. Install substrate board with long joints in continuous straight lines, perpendicular to roof slopes with end joints staggered between rows. Tightly butt substrate boards together.
 - 1. Fasten substrate board to top flanges of steel deck according to recommendations in FM Approvals' "RoofNav" and FM Global Loss Prevention Data Sheet 1-29 for specified Windstorm Resistance Classification.
 - 2. Fasten substrate board to top flanges of steel deck to resist uplift pressure at corners, perimeter, and field of roof according to membrane roofing system manufacturers' written instructions.

3.4 PROTECTION BOARD

- A. Install substrate board over rigid insulation with long joints in continuous straight lines, perpendicular to roof slopes with end joints staggered between rows. Tightly butt substrate boards together.

3.5 VAPOR-RETARDER INSTALLATION

- A. Polyethylene Film: Loosely lay polyethylene-film vapor retarder in a single layer over area to receive vapor retarder, side and end lapping each sheet a minimum of 2 inches and 6 inches, respectively.
 - 1. Continuously seal side and end laps with tape.
- B. Completely seal vapor retarder at terminations, obstructions, and penetrations to prevent air movement into membrane roofing system.
- C. Coordinate with roofing manufacturer location of vapor retarder.

3.6 INSULATION INSTALLATION

- A. Coordinate installing membrane roofing system components so insulation is not exposed to precipitation or left exposed at the end of the workday.
- B. Comply with membrane roofing system and insulation manufacturer's written instructions for installing roof insulation.
- C. Install insulation under area of roofing to achieve required thickness. Where overall insulation thickness is 2.7 inches or greater, install two or more layers with joints of each succeeding layer staggered from joints of previous layer a minimum of 6 inches in each direction.
- D. Trim surface of insulation where necessary at roof drains so completed surface is flush and does not restrict flow of water.
- E. Install insulation with long joints of insulation in a continuous straight line with end joints staggered between rows, abutting edges and ends between boards. Fill gaps exceeding 1/4 inch with insulation.
 - 1. Cut and fit insulation within 1/4 inch of nailers, projections, and penetrations.
- F. Mechanically Fastened Insulation: Install each layer of insulation and secure to deck using mechanical fasteners specifically designed and sized for fastening specified board-type roof insulation to deck type.
 - 1. Fasten insulation according to requirements in FM Approvals' "RoofNav" for specified Windstorm Resistance Classification.
 - 2. Fasten insulation to resist uplift pressure at corners, perimeter, and field of roof.

3.7 MECHANICALLY FASTENED MEMBRANE ROOFING INSTALLATION

- A. Mechanically fasten membrane roofing over area to receive roofing and install according to roofing system manufacturer's written instructions.
 - 1. For in-splice attachment, install membranes roofing with long dimension perpendicular to steel roof deck flutes.
- B. Start installation of membrane roofing in presence of roofing system manufacturer's technical personnel.
- C. Accurately align membrane roofing and maintain uniform side and end laps of minimum dimensions required by manufacturer. Stagger end laps.
- D. Mechanically fasten or adhere membrane roofing securely at terminations, penetrations, and perimeter of roofing.
- E. Apply membrane roofing with side laps shingled with slope of roof deck where possible.
- F. In-Seam Attachment: Secure one edge of TPO sheet using fastening plates or metal battens centered within membrane seam and mechanically fasten TPO sheet to roof deck.
- G. Seams: Clean seam areas, overlap membrane roofing, and hot-air weld side and end laps of membrane roofing and sheet flashings according to manufacturer's written instructions to ensure a watertight seam installation.
 - 1. Test lap edges with probe to verify seam weld continuity. Apply lap sealant to seal cut edges of sheet membrane.
 - 2. Verify field strength of seams a minimum of twice daily and repair seam sample areas.
 - 3. Repair tears, voids, and lapped seams in roofing that does not comply with requirements.
- H. Spread sealant bed over deck drain flange at roof drains and securely seal membrane roofing in place with clamping ring.
- I. Install membrane roofing and auxiliary materials to tie in to existing roofing to maintain weathertightness of transition.

3.8 BASE FLASHING INSTALLATION

- A. Install sheet flashings and preformed flashing accessories and adhere to substrates according to membrane roofing system manufacturer's written instructions.
- B. Apply bonding adhesive to substrate and underside of sheet flashing at required rate and allow to partially dry. Do not apply to seam area of flashing.
- C. Flash penetrations and field-formed inside and outside corners with cured or uncured sheet flashing.
- D. Clean seam areas, overlap, and firmly roll sheet flashings into the adhesive. Hot-air weld side and end laps to ensure a watertight seam installation.

- E. Terminate and seal top of sheet flashings and mechanically anchor to substrate through termination bars.

3.9 WALKWAY INSTALLATION

- A. Flexible Walkways: Install walkway products in locations indicated. Heat weld to substrate or adhere walkway products to substrate with compatible adhesive according to roofing system manufacturer's written instructions.

3.10 FIELD QUALITY CONTROL

- A. Installation Inspections: Engage a full-time field quality control inspector during roof system installation. The field quality control inspector shall submit inspection report at weekly intervals to Contractor and COR and coordinate with roofing system manufacturer's technical personnel and inspect roofing installation upon completion.
- B. Testing Agency: Engage a qualified testing agency to inspect substrate conditions, surface preparation, membrane application, flashings, protection, and drainage components, and to furnish reports to COR.
 - 1. Electric Field Vector Mapping (EFVM): Testing agency shall survey entire roof area for potential leaks using electric field vector mapping (EFVM).
- C. Final Roof Inspection: Arrange for roofing system manufacturer's technical personnel to inspect roofing installation on completion.
- D. Repair or remove and replace components of membrane roofing system where inspections indicate that they do not comply with specified requirements.
- E. Additional inspections, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.

3.11 PROTECTING AND CLEANING

- A. Protect membrane roofing system from damage and wear during remainder of construction period. When remaining construction will not affect or endanger roofing, inspect roofing for deterioration and damage, describing its nature and extent in a written report, with copies to COR.
- B. Correct deficiencies in or remove membrane roofing system that does not comply with requirements; repair substrates; and repair or reinstall membrane roofing system to a condition free of damage and deterioration at time of Substantial Completion and according to warranty requirements.
- C. Clean overspray and spillage from adjacent construction using cleaning agents and procedures recommended by manufacturer of affected construction.

END OF SECTION 07 54 23

SECTION 07 72 00 - ROOF ACCESSORIES

PART 1 - GENERAL

1.1 SUMMARY

A. This Section includes the following:

1. Manufacturers.
2. Metal Materials
3. Roof Hatches.

1.2 SUBMITTALS

- A. Product Data: For each type of roof accessory indicated. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.
- B. Shop Drawings: Show fabrication and installation details for roof accessories. Show layouts of roof accessories including plans and elevations. Indicate dimensions, weights, loadings, required clearances, method of field assembly, and components. Include plans, elevations, sections, details, and attachments to other work.
- C. Coordination Drawings: Roof plans, drawn to scale, and coordinating penetrations and roof-mounted items. Show the following:
1. Size and location of roof accessories specified in this Section.
 2. Method of attaching roof accessories to roof or building structure.
 3. Other roof-mounted items including mechanical and electrical equipment, ductwork, piping, and conduit.
- D. Samples: For each type of exposed factory-applied[**color**] finish required and for each type of roof accessory indicated, prepared on Samples of size to adequately show color.
- E. Warranty: Special warranty specified in this Section.

1.3 QUALITY ASSURANCE

- A. Sheet Metal Standard: Comply with SMACNA's "Architectural Sheet Metal Manual" details for fabrication of units, including flanges and cap flashing to coordinate with type of roofing indicated.

1.4 DELIVERY, STORAGE, AND HANDLING

- A. Pack, handle, and ship roof accessories properly labeled in heavy-duty packaging to prevent damage.

1.5 PROJECT CONDITIONS

- A. Field Measurements: Verify required openings for each type of roof accessory by field measurements before fabrication and indicate measurements on Shop Drawings.

1.6 COORDINATION

- A. Coordinate layout and installation of roof accessories with **[roofing membrane and base flashing and]**interfacing and adjoining construction to provide leakproof, weathertight, secure, and noncorrosive installation.
 - 1. With COTR's approval, adjust location of roof accessories that would interrupt **[roof drainage routes] [roof expansion joints] <Insert construction element>**.

1.7 WARRANTY

- A. Special Warranty on Painted Finishes: Manufacturer's standard form in which manufacturer agrees to repair finish or replace roof accessories that show evidence of deterioration of factory-applied finishes within specified warranty period.
 - 1. Fluoropolymer Finish: Deterioration includes, but is not limited to, the following:
 - a. Color fading more than 5 Hunter units when tested according to ASTM D 2244.
 - b. Chalking in excess of a No. 8 rating when tested according to ASTM D 4214.
 - c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.
 - 2. Finish Warranty Period: **[20] <Insert number>** years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, manufacturers listed in other Part 2 articles.
- B. Manufacturers: Subject to compliance with requirements, provide products by one of the manufacturers listed in other Part 2 articles.

2.2 METAL MATERIALS

- A. Galvanized Steel Sheet: ASTM A 653/A 653M, G90 coated[**and mill phosphatized for field painting**].
- B. Aluminum-Zinc Alloy-Coated Steel Sheet: ASTM A 792/A 792M, AZ50 coated.

- C. Prepainted, Metallic-Coated Steel Sheet: Steel sheet metallic coated by hot-dip process and prepainted by coil-coating process to comply with ASTM A 755/A 755M.
1. Galvanized Steel Sheet: ASTM A 653/A 653M, G90 coated.
 2. Aluminum-Zinc Alloy-Coated Steel Sheet: ASTM A 792/A 792M, Class AZ50 coated.
 3. Exposed Finishes: High-Performance Organic Finish (2-Coat Fluoropolymer): Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturer's written instructions.
 - a. Fluoropolymer 2-Coat System: Manufacturer's standard 2-coat, thermocured system consisting of specially formulated inhibitive primer and fluoropolymer color topcoat containing not less than 70 percent polyvinylidene fluoride resin by weight; complying with physical properties and coating performance requirements in [AAMA 2604] [AAMA 2605], except as modified below:
 - 1) Humidity Resistance: [1000] [2000] hours.
 - 2) Salt-Spray Resistance: [1000] [2000] hours.
- D. Aluminum Sheet: ASTM B 209, alloy and temper recommended by manufacturer for type of use and [mill] finish. [**Coil-coat finish as follows:**]
1. Factory-Prime Coating: Where painting after installation is indicated, provide pretreatment and white or light-colored, factory-applied, baked-on epoxy primer coat; with a minimum dry film thickness of 0.2 mil.
 2. Class II, Clear Anodic Finish: AA-M12C22A31 (Mechanical Finish: Nonspecular as fabricated; Chemical Finish: Etched, medium matte; Anodic Coating: Architectural Class II, clear coating 0.010 mm or thicker) complying with AAMA 611.
 3. Class II, Color Anodic Finish: AA-M12C22A32/A34 (Mechanical Finish: Nonspecular as fabricated; Chemical Finish: Etched, medium matte; Anodic Coating: Architectural Class II, integrally colored or electrolytically deposited color coating 0.010 mm or thicker) complying with AAMA 611.
 - a. Color: [Light bronze] [Medium bronze] [Dark bronze] [Black] [Champagne] <Insert color>.
 4. Baked-Enamel Finish: AA-C12C42R1x (Chemical Finish: Cleaned with inhibited chemicals; Chemical Finish: Acid-chromate-fluoride-phosphate conversion coating; Organic Coating: As specified below). Apply baked enamel complying with paint manufacturer's written instructions for cleaning, conversion coating, and painting.
 - a. Organic Coating: Thermosetting, modified-acrylic enamel primer/topcoat system complying with AAMA 2603 except with a minimum dry film thickness of 1.5 mils, medium gloss.
 - b. Color and Gloss: [As indicated by manufacturer's designations] [Match Architect's sample] [As selected by Architect from manufacturer's full range] <Insert color and gloss>.
 5. High-Performance Organic Finish (2-Coat Fluoropolymer): AA-C12C40R1x (Chemical Finish: Cleaned with inhibited chemicals; Chemical Finish: Conversion coating; Organic Coating: Manufacturer's standard 2-coat, thermocured system consisting of specially formulated inhibitive primer and fluoropolymer color topcoat containing not

less than 70 percent polyvinylidene fluoride resin by weight). Prepare, pretreat, and apply coating to exposed metal surfaces to comply with [AAMA 2604] [AAMA 2605] and with coating and resin manufacturer's written instructions.

- a. Color and Gloss: [As indicated by manufacturer's designations] [Match Architect's sample] [As selected by Architect from manufacturer's full range] <Insert color and gloss>.
6. Powder-Coat Finish: Immediately after cleaning and pretreating, electrostatically apply manufacturer's standard baked-polymer thermosetting powder finish. Comply with resin manufacturer's written instructions for application, baking, and minimum dry film thickness.
- a. Color and Gloss: [As indicated by manufacturer's designations] [Match Architect's sample] [As selected by Architect from manufacturer's full range] <Insert color and gloss>.
- E. Aluminum Extrusions and Tubes: ASTM B 221, alloy and temper recommended by manufacturer for type of use, mill finished.
- F. Stainless-Steel Shapes or Sheet: ASTM A 240/A 240M or ASTM A 666, Type 304 or Type 316, No. 2D finish.
- G. Steel Shapes: ASTM A 36/A 36M, hot-dip galvanized to comply with ASTM A 123/A 123M, unless otherwise indicated.
- H. Steel Tube: ASTM A 500, round tube, baked-enamel finished.
- I. Galvanized Steel Tube: ASTM A 500, round tube, hot-dip galvanized to comply with ASTM A 123/A 123M.
- J. Galvanized Steel Pipe: ASTM A 53/A 53M.
1. .

2.3 ROOF HATCHES

- A. Roof Hatches: Fabricate roof hatches with insulated double-wall lids and insulated [single] [double]-wall curb frame with integral deck mounting flange and lid frame counterflashing. Fabricate with welded or mechanically fastened and sealed corner joints. Provide continuous weathertight perimeter gasketing and equip with corrosion-resistant or hot-dip galvanized hardware.
1. [Available]Manufacturers:
 - a. Babcock-Davis; a Cierra Products Inc. Company.
 - b. Bilco Company (The).
 - c. Bristolite Skylights.
 - d. Custom Curb, Inc.
 - e. Dur-Red Products.
 - f. Hi Pro International, Inc.

- g. J. L. Industries, Inc.
 - h. Metallic Products Corporation.
 - i. Milcor Inc.; a Gibraltar Company.
 - j. Nystrom, Inc.
 - k. O'Keeffe's Inc.
 - l. Precision Ladders, LLC.
 - m. Roof Products & Systems Corporation.
 - n. ThyCurb; Div of Thybar Corporation.
 - o. Wasco Products, Inc.
 - p. Western Canwell.
 - q. **<Insert manufacturer's name.>**
- 2. Loads: Fabricate roof hatches to withstand **[40-lbf/sq. ft.] <Insert load>** external and **[20-lbf/sq. ft.] <Insert load>** internal loads.
 - 3. Type and Size: Single-leaf lid, **[30 by 36 inches] [30 by 54 inches] [30 by 96 inches] <Insert size>**.
 - 4. Type and Size: Double-leaf lid, **[72 by 96 inches] <Insert size>**.
 - 5. Curb and Lid Material: **[Galvanized] [Aluminum-zinc alloy-coated]** steel sheet, **[0.079 inch] <Insert thickness>** thick.
 - 6. Curb and Lid Material: Aluminum sheet, **[0.090 inch] <Insert thickness>** thick.
 - 7. Curb and Lid Material: Stainless-steel sheet, **[0.078 inch] <Insert thickness>** thick.
 - a. Finish: **[Prime painted] [Baked enamel] [High-performance organic coating] [Powder coat]**.
 - b. Finish: **[Mill] [Clear anodic] [Color anodic]**.
 - 8. Insulation: **[Cellulosic-fiber] [Glass-fiber] [Polyisocyanurate]** board.
 - 9. Interior Lid Liner: Manufacturer's standard metal liner of same material and finish as outer metal lid.
 - 10. Exterior Curb Liner: Manufacturer's standard metal liner of same material and finish as metal curb.
 - 11. On ribbed or fluted metal roofs, form flange at perimeter bottom to conform to roof profile.
 - 12. Fabricate units to minimum height of **[12 inches] <Insert height>**, unless otherwise indicated.
 - 13. Sloping Roofs: Where slope or roof deck exceeds 1:48, fabricate hatch curbs with height **[constant] [tapered to match slope to level tops of units]**.
 - 14. Hardware: **[Galvanized steel] [Stainless-steel]** spring latch with turn handles, butt- or pintle-type hinge system, and padlock hasps inside and outside.
 - a. Provide 2-point latch on covers larger than 84 inches.
 - b. Provide remote-control operation.
 - 15. Ladder Safety Post: Manufacturer's standard ladder safety post. Post to lock in place on full extension. Provide release mechanism to return post to closed position.
 - a. Test Load: **<Insert load.>**
 - b. Height: **[42 inches] <Insert height>** above finished roof deck.
 - c. Material and Finish: **[Steel tube, galvanized] [Steel tube, baked enameled] [Stainless steel, mill finished] [Aluminum, mill finished]**.
 - d. Diameter: Pipe with **[1-5/8-inch] <Insert diameter>** OD tube.

16. Safety Railing System: Manufacturer's standard complete system including rails, clamps, fasteners, safety barrier at railing opening, and all accessories required for a complete installation.
 - a. Test Load: **<Insert load.>**
 - b. Height: **[42 inches]** **<Insert height>** above finished roof deck.
 - c. Pipe or Tube: 1-1/4-inch ID galvanized pipe or 1-5/8-inch OD galvanized tube.
 - d. Flat Bar: 2-inch- high by 3/8-inch- thick galvanized steel.
 - e. Chain Passway Enclosure: Galvanized proof coil chain with quick link on fixed end.
 - f. Self-Latching Gate: Fabricated of same materials and rail spacing as safety railing system. Provide manufacturer's standard hinges and self-latching mechanism.
 - g. Pipe Ends and Tops: Covered or plugged with weather-resistant material.
 - h. Provide weep holes or another means to drain entrapped water in hollow sections of handrail and railing members that are exposed to exterior or to moisture from condensation or other sources.
 - i. Fabricate joints that will be exposed to weather in a watertight manner.
 - j. Close exposed ends of handrail and railing members with prefabricated end fittings.
 - k. Fasteners: Manufacturer's standard.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine substrates, areas, and conditions, with Installer present, to verify actual locations, dimensions, and other conditions affecting performance of work.
 1. Verify that substrate is sound, dry, smooth, clean, sloped for drainage, and securely anchored and is ready to receive roof accessories.
 2. Verify dimensions of roof openings for roof accessories.
 3. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

- A. General: Install roof accessories according to manufacturer's written instructions. Anchor roof accessories securely in place and capable of resisting forces specified. Use fasteners, separators, sealants, and other miscellaneous items as required for completing roof accessory installation. Install roof accessories to resist exposure to weather without failing, rattling, leaking, and fastener disengagement.
- B. Install roof accessories to fit substrates and to result in watertight performance.
- C. Metal Protection: Where dissimilar metals will contact each other or corrosive substrates, protect against galvanic action by painting contact surfaces with bituminous coating or by other permanent separation as recommended by manufacturer.

1. Coat concealed side of [**uncoated aluminum**] [**stainless-steel**] roof accessories with bituminous coating where in contact with wood, ferrous metal, or cementitious construction.
 2. Underlayment: Where installing exposed-to-view components of roof accessories directly on cementitious or wood substrates, install a course of felt underlayment and cover with a slip sheet, or install a course of polyethylene underlayment.
 3. Bed flanges in thick coat of asphalt roofing cement where required by roof accessory manufacturers for waterproof performance.
- D. Install roof accessories level, plumb, true to line and elevation, and without warping, jogs in alignment, excessive oil canning, buckling, or tool marks.
- E. Roof Curb Installation:
1. Set roof curb so top surface of roof curb is level.
- F. Equipment Support Installation:
1. Set equipment support so top surface of equipment support is level.
- G. Roof Hatch Installation:
1. Check roof hatch for proper operation. Adjust operating mechanism as required. Clean and lubricate joints and hardware.
 2. Attach safety railing system to roof hatch curb.
 3. Attach ladder safety post according to manufacturer's written instructions.
- H. Heat and Smoke Vent Installation: Locate, install, and test heat and smoke vents according to NFPA 204.
1. Check heat and smoke vent for proper operation. Adjust operating mechanism as required.
- I. Gravity Ventilator Installation:
1. Check relief vents for proper operation and unrestricted airflow.
 2. Install security grilles. Weld bar intersections and ends of bars to structural frame or primary curb walls.
- J. Roof Walkway Installation:
1. Verify location of points of access to roof-mounted equipment via use of roof walkways.
 2. Remove ballast from top surface of low-slope roofing section surface at the full area of contact surface of roof walkway supports.
 3. Verify that roof walkway support isolation pads are in place prior to placement of roof walkway onto low-slope roofing section surface.
- K. Preformed Flashing Installation:
1. Secure to roof membrane according to vent and stack flashing manufacturer's written instructions.

- L. Seal joints with [**elastomeric**] [**butyl**] sealant as required by manufacturer of roof accessories.

3.3 TOUCH UP

- A. Touch up factory-primed surfaces with compatible primer ready for field painting in accordance with Division 09 painting Sections.
- B. Galvanized Surfaces: Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A 780.

3.4 CLEANING

- A. Clean exposed surfaces according to manufacturer's written instructions.

END OF SECTION 07 72 00